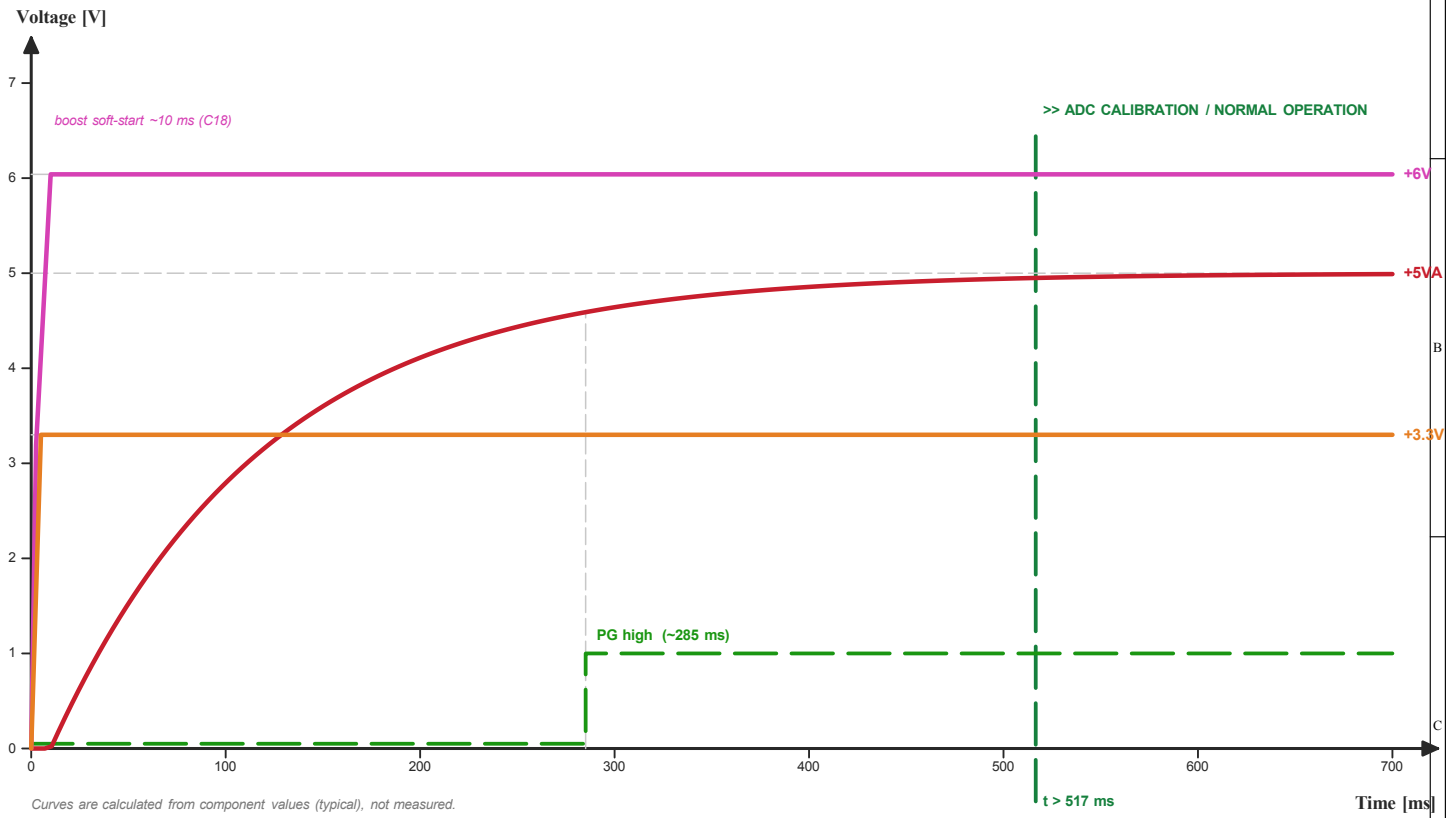
[9] Power - Sequencing

NAME	SOURCE	LEVEL
+3.3V	Host (J1)	3.3 V (host)
+6V	TPS61086 IC2	6.04 V
+5VA	LT3042 U2	5.00 V
PG	LT3042 U2	> 4.59 V

Values from the schematic:
+6V = 1.238 V x (1 + R17/R15) = 1.238 x (1 + 69.8k/18k)
+5VA = ISET x RSET = 100 uA x 50k (R9)
PG threshold = 0.3 V x (1 + R12/R13) = 4.59 V
LT3042 soft-start: tau = RSET x CSET = 50k x 2.2 uF
= 110 ms -> +5VA at 99 % after ~550 ms

DESIGN NOTE:
The LT3042 SET capacitor (C14 = 2.2 uF) gives a 110 ms time constant. Firmware must wait >= 600 ms after power-up (or until PG, TP17) before the AD7190 reset and internal calibration. To start faster, reduce C14 (e.g. 0.47 uF -> tau = 23.5 ms) at the cost of slightly more noise.



Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: 349b121
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 9 of 10